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CS23532-COMPUTER NETWORKS-LAB MANUAL

PRACTICAL – 10 (a) , 10 (b) & 10 (c)

AIM:

To design and configure internetworks using routers, switches, wireless routers, DHCP servers, Internet cloud, and Ethernet cables in Cisco Packet Tracer, assign IP addresses, establish communication between different networks, and observe packet routing during transmission.

PART (a) : Design and Configure a Simple Internetwork Using a Router

1. **Design the Network:**
   Create two different networks, each containing 2–3 PCs, and connect them through a router.

- 1 Router — Cisco 2911
- 2 Switches — Cisco 2960
- 4 PCs
- Copper Straight-Through cables

2. **Assign IP Addresses:**
   - Use two different networks.
   - **Network 1:**
     Network: `192.168.1.0/24`
   - **Network 2:**
     Network: `192.168.2.0/24`

3. **Configure Router0:**
   OPEN:
   Router0 → CLI
   Command Prompt:

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

end
show ip interface brief
```

Outcome:

```
up    up
```

4. **Test Communication:**

   PC0 → Desktop → Command Prompt

   Command:

   ```
   ping 192.168.2.10
   ```

   Outcome:

   ```
   Packets: Sent = 4
   Received = 4
   Lost = 0
   ```

5. **Simulate and Trace Packet Routing:**

   - Select **Simulation** mode.

- Select **Add Simple PDU**.
- Click **PC0**.
- Click **PC2**.
- Click **Auto Capture / Play**.
- Observe the packet travelling through the switches and router.

  PC0 → Switch0 → Router0 → Switch1 → PC2

PART (b) : Internetwork Using Wireless Router, DHCP Server and Internet Cloud

1. **Design the Wireless Network:**
   - 1 Wireless Router
   - 2 Wireless PCs/Laptops
   - 1 Server
   - 1 Cloud
   - Ethernet cables where required

2. **Configure the Wireless Router:**

   Configure the LAN:

   IP Address: 192.168.10.1

   Subnet Mask: 255.255.255.0

   Enable :

   DHCP Server: ON

   Set the DHCP range:

   Starting IP: 192.168.10.100

   Maximum Users: 50

3. **Configure Wireless Devices:**

   Connect the laptops to the wireless network.

Set their IP configuration to: DHCP

4. **Configure the Internet Cloud:**

   - Connect the wireless router to the **Internet Cloud** using an Ethernet connection.
   - Observe the communication between the internal network and the Internet cloud.

PART (c) : Internetwork Using Switch, Router and Ethernet Cables

1. **Design the Network:**

   - 1 Router
   - 2 Switches
   - 4 PCs
   - Ethernet cables

2. **Assign IP Addresses:**

   - **First Network:**
     PC0 → 10.0.0.10
     PC1 → 10.0.0.20
     Router G0/0 → 10.0.0.1

   - **Second Network:**
     PC2 → 10.0.1.10
     PC3 → 10.0.1.20
     Router G0/1 → 10.0.1.1

   - **Subnet Mask:**

255.255.255.0

- **Default gateways:**
  PC0, PC1 → 10.0.0.1
  PC2, PC3 → 10.0.1.1

3. **Configure the Router:**

- **Command Prompt:**
  enable
  configure terminal

  interface gigabitEthernet 0/0
  ip address 10.0.0.1 255.255.255.0
  no shutdown
  exit

  interface gigabitEthernet 0/1
  ip address 10.0.1.1 255.255.255.0
  no shutdown
  exit

  end

- **Verify:**
  show ip interface brief

4. **Test Communication:**

- **From PC0:**
  ping 10.0.1.10

Successful communication confirms that the router is forwarding packets between the two networks.

RESULT:

Thus, the experiment successfully demonstrated internetworking, static IP addressing, router-based packet forwarding, DHCP, wireless networking, and Ethernet-based communication.